Q1

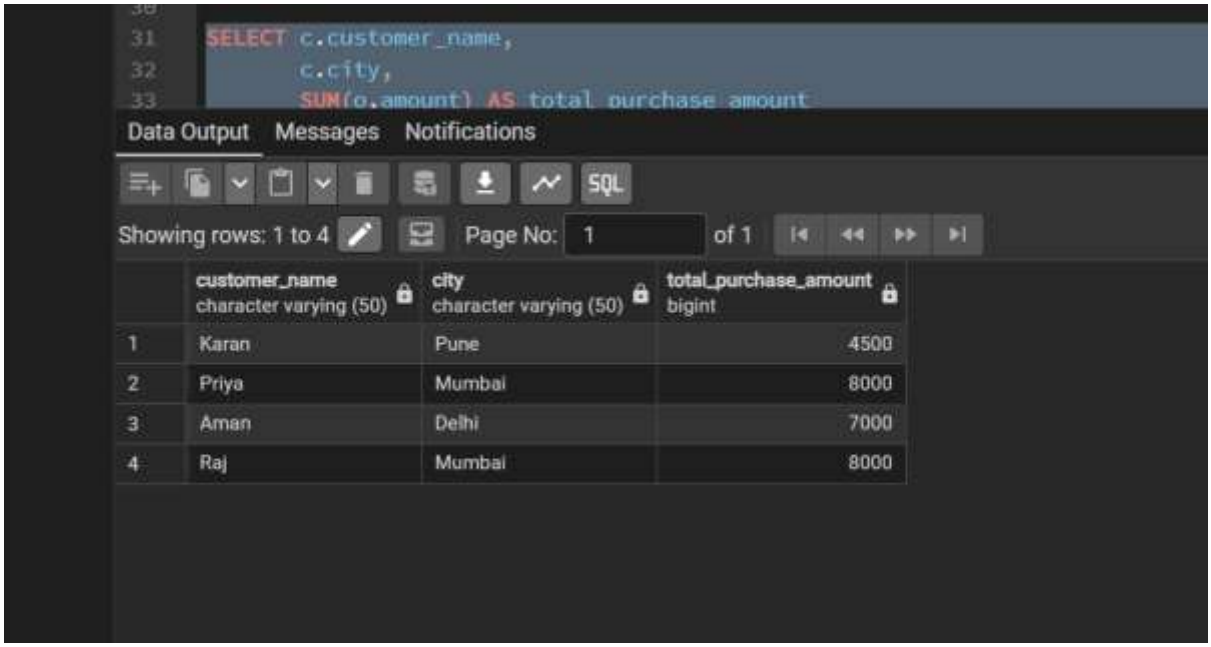
```
CREATE TABLE Customer (
   customer_id INT PRIMARY KEY,
   customer_name VARCHAR(50),
   city VARCHAR(50)
);

CREATE TABLE Orders1 (
   order_id INT PRIMARY KEY,
   customer_id INT REFERENCES Customer(customer_id),
   amount INT
);
INSERT INTO Customer (customer_id, customer_name, city)
VALUES
(1, 'Aman', 'Delhi'),
(2, 'Neha', 'Delhi'),

(3, 'Raj', 'Mumbai'),
(4, 'Priya', 'Mumbai'),
(5, 'Karan', 'Pune');

INSERT INTO Orders1 (order_id, customer_id, amount)
VALUES
(101, 1, 4000),
(102, 1, 3000),
(103, 2, 5000),
(104, 3, 6000),
(105, 3, 2000),
(106, 4, 8000),
(107, 5, 4500);
```

```
SELECT c.customer_name,
    c.city,
    SUM(o.amount) AS total_purchase_amount
FROM Customer c
JOIN Orders o
  ON c.customer_id = o.customer_id
GROUP BY c.customer_id, c.customer_name, c.city
HAVING NOT EXISTS (
  SELECT 1
  FROM Customer c2
  JOIN Orders o2
    ON c2.customer_id = o2.customer_id
  WHERE c2.city = c.city
  GROUP BY c2.customer_id
  HAVING SUM(o2.amount) > SUM(o.amount)
);
```



	customer_name character varying (50)	city character varying (50)	total_purchase_amount bigint
1	Karan	Pune	4500
2	Priya	Mumbai	8000
3	Aman	Delhi	7000
4	Raj	Mumbai	8000

Q2:

```
CREATE TABLE product_inventory (
   product_id INT PRIMARY KEY,
   product_name VARCHAR(50),
   price INT
);

CREATE TABLE product_audit (
   product_id INT,
   product_name VARCHAR(50),
   action VARCHAR(30),
   action_time TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE OR REPLACE FUNCTION log_product_changes()
RETURNS TRIGGER
LANGUAGE plpgsql
AS $$
BEGIN
```

```
  IF TG_OP = 'INSERT' THEN
    INSERT INTO product_audit
    VALUES (
      NEW.product_id,
      NEW.product_name,
      'PRODUCT ADDED',
      CURRENT_TIMESTAMP
    );

  ELSIF TG_OP = 'DELETE' THEN
    INSERT INTO product_audit
    VALUES (
      OLD.product_id,
      OLD.product_name,
      'PRODUCT REMOVED',
      CURRENT_TIMESTAMP
    );
  END IF;

  RETURN NULL;
END;
$$;

CREATE TRIGGER product_audit_trigger
AFTER INSERT OR DELETE
ON product_inventory
FOR EACH ROW
EXECUTE FUNCTION log_product_changes();

INSERT INTO product_inventory
VALUES (1, 'Laptop', 50000);
```

```
DELETE FROM product_inventory
WHERE product_id = 1;
```

```
SELECT * FROM product_audit;
```

